

A Belief-Engine Decision Model for Normalizing Scraped E-Commerce Product Specifications

From belief and cost to *information*: which clue to buy next, and when to stop — a thinking-first prototype that decides, per field, whether a value is safe to publish

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Abstract

Scraping product data is the easy part; deciding whether a scraped value is *correct* is the hard part, because the true canonical specification is never observed at decision time. We model each scraped field (e.g. `ram_gb`, `storage_gb`, `weight_g`) as one of six mutually-exclusive hidden “worlds” (`correct`, `unit_error`, `wrong_product`, `typo`, `missing`, `garbled`), start from a source-reliability prior, update it with cheap evidence signals via Bayes’ rule, and choose among ACCEPT, REPAIR, BUY-MORE-EVIDENCE, FLAG-FOR-HUMAN, and REJECT by *minimum expected loss* under an asymmetric cost matrix in which publishing a wrong value is by far the most expensive outcome. We evaluate on 40 labelled cases (labels hidden at decision time) against an autonomous no-human variant and a deliberately “dumb” score-plus-threshold baseline. On this small sample the full policy attains the lowest average decision cost (2.30 vs. 3.10 for the baseline; paired Wilcoxon $p \approx 0.015$), but with three important honesty caveats: (i) the win is *narrow*—the median per-case cost difference is 0 and only 9 of 40 cases differ; (ii) all three policies commit the same two catastrophic wrong-publishes, so the belief layer buys its lower average cost mostly by escalating 57.5% of cases to a human rather than by preventing the expensive errors; and (iii) the model’s top confidence bin is empirically only $\sim 60\%$ correct, which—if real—would invert the very ACCEPT decision the expected-loss rule is meant to make. We therefore present this as a *calibrated way of thinking* about the problem, not a production system, and we are explicit about which numbers are assumptions and which are measured. We then add an *information layer*: we measure the agent’s doubt in *bits* (Shannon entropy), rank candidate evidence by *expected information gain per unit cost*, and replace the earlier flat two-buy cap with a true *value-of-information* (VOI) stop rule. Three findings stand out and are honest about their own limits: (a) a clue can flip which world leads while barely reducing entropy (the leading out-of-support signal cuts doubt by only 0.24 of 2.12 bits); (b) the most *informative* clue is not the best *buy* once cost enters (the manufacturer proxy leads on raw bits but a cheaper cross-source panel wins on bits-per-cost); and (c) under our own cost matrix *every* candidate clue has *zero* value of information—the guaranteed human-review cost (FLAG= 2) is such a strong ceiling that the agent should flag, not buy, and we show by prediction-and-test that even making the human much more expensive does *not* rescue buying on this high-entropy case (VOI plateaus below the clue cost), because no single clue can concentrate belief enough to escape into a cheaper action. All priors, likelihoods, and costs are author-stated assumptions with recorded provenance; three adversarial AI reviews and five human discussions shaped the design and are documented alongside.

Contents

1	Introduction and problem statement	3
2	Related work and positioning	3
3	Hidden states: six MECE worlds	4
4	The belief model	4
4.1	Source-reliability prior	4
4.2	Likelihoods	5
4.3	Bayesian update and chaining	5
5	Assumptions, provenance, and the cost matrix	5
5.1	What is assumed vs. measured	5
5.2	Asymmetric cost matrix	6
6	Policy: minimum expected loss with an abstain option	6
6.1	Decision rule	6
6.2	Derived ACCEPT threshold (Chow’s reject option)	6
6.3	Buying information: a budget heuristic, not true VOI	6
7	The information layer: bits, which clue to buy, and when to stop	7
7.1	Measuring doubt in bits	7
7.2	Expected vs. observed gain	7
7.3	Which clue to buy next: expected bits per unit cost	8
7.4	When to stop: a true value-of-information rule	8
7.5	Two intuitions, tested honestly	9
7.6	What the information layer exposed (a falsified hypothesis)	9
8	Experimental setup (methods)	9
9	Results	10
9.1	Headline metrics	10
9.2	Cost advantage: significant but narrow	10
9.3	Calibration	10
9.4	Escalation and the shared blind spot	11
10	Does the belief layer “earn its keep”?	12
11	Failure modes and limitations	12
12	Threats to validity	13
13	AI-use statement, human discussions, and conclusion	13
A	Required core questions	14

1 Introduction and problem statement

E-commerce catalogs are assembled from scraped and syndicated data: the same phone’s specifications are pulled from manufacturer pages, retailer listings, and price aggregators, then normalized into one record. Scraping itself is largely solved; the expensive, unglamorous problem is deciding whether a scraped *value* is trustworthy enough to publish. A field can be perfectly parseable and still wrong: 16000 MB of RAM (a unit slip), 160 GB of RAM (an impossible typo), 512 GB of storage that is real but belongs to a different variant, or a numeric value that five sites agree on only because they all copied the same bad upstream feed.

The defining difficulty is that **the true canonical specification is never observed at decision time**. The agent sees clues—the value, its plausibility, what other sources say—but never the ground truth. This is a decision-under-uncertainty problem, not a classification problem, and the right tool is a *belief* over what might have happened, combined with a *cost* of being wrong.

The decision. For each scraped field the agent must choose one of:

- ACCEPT — publish the value as-is;
- REPAIR — transform then publish (e.g. 16000 MB $\rightarrow$ 16 GB);
- BUY — pay to gather more evidence (re-scrape, query a manufacturer proxy) before deciding;
- FLAG — send to a human reviewer;
- REJECT — drop the value (publish nothing for this field).

Scope. We work in consumer electronics (phones/laptops), one decision per scraped field. We assume the upstream step correctly identified *which* field a value belongs to; schema/key-mismatch across sources is out of scope and revisited as a limitation (§11).

Contributions and honest framing. This is a thinking-first prototype combining two deliverables. Its contribution is a *calibrated way of thinking*: an explicit, auditable pipeline from source-reliability prior, through Bayesian updates on cheap signals, to a minimum-expected-loss decision with a human escape hatch, *extended with an information layer* that measures doubt in bits, ranks evidence by expected information gain per unit cost, and decides when to stop buying by true value of information (§7). Every prior, likelihood, and cost is an author-stated assumption with recorded provenance (§5); we never present a guess as a measured fact. The empirical results (§9) are deliberately reported with their caveats: a small sample, a cost advantage that is statistically real but *narrow*, and a calibration problem in the highest-stakes confidence bin. The information analysis adds its own honest twist—under our cost matrix, most doubtful cases have zero value of information, so the agent should flag rather than buy, revealing that the buy policy was silently driven by an optimistic flat human-review cost.

2 Related work and positioning

The components here are standard; the contribution is their explicit, honest assembly for a concrete data-quality task.

- **Bayesian decision theory.** Choosing the action with minimum expected loss under a posterior is the textbook Bayes decision rule [1], and the prior $\rightarrow$ likelihood $\rightarrow$ posterior update with weakly-informative priors follows standard practice [4].

- **Reject-option classification.** Our “act only above a cost-derived confidence, otherwise abstain to a human” rule is exactly Chow’s reject option [2]: the FLAG action is a principled abstain, not an admission of defeat.
- **Cost-sensitive learning.** The asymmetric cost matrix and the insight that one must not simply pick the most-likely class under asymmetric losses is Elkan’s cost-sensitive framing [3].
- **Calibration.** That predicted probabilities must match empirical frequencies—and that modern models are often overconfident—motivates our calibration analysis and the recalibration remedies (Platt/temperature/isotonic scaling) [5, 6, 11].
- **Sequential testing / value of information.** The “buy evidence then re-decide” loop is, in its fully-correct form, a sequential decision problem [9]; we use a budget heuristic and say so (§6).
- **Active learning and product matching.** Our human-in-the-loop routing is an active-learning setup [8], which biases evaluation (§12); the `wrong_product` problem connects to large-scale product matching benchmarks such as WDC Products [7].

Where an idea came from an online discussion or the course rather than a paper, we say so in-text rather than manufacture a citation.

3 Hidden states: six MECE worlds

The hidden reality behind a value is exactly one of six worlds, chosen to be mutually exclusive and collectively exhaustive (MECE). A tie-breaker, applied in order, guarantees a value never sits in two boxes: (1) unparseable as this field → **garbled**; (2) empty/placeholder → **missing**; (3) parses to a number → decide among the remaining worlds using the plausibility (“support”) and unit-conversion tests.

World	The reality that produced the value
correct	Value matches the true canonical spec for this product+field.
unit_error	Right magnitude, wrong unit (16000MB → 16 GB). Narrow by design; a broader class of <i>measurement-definition</i> errors (net vs. gross weight, diagonal vs. width) is noted as future work.
wrong_product	A valid, in-support value taken from a different product/variant/revision (includes <i>stale</i>). Looks normal in isolation.
typo	Transcription slip producing an off/out-of-support number (160 GB RAM).
missing	Empty/null/placeholder-empty.
garbled	Present but unparseable as this field (encoding junk, HTML, “N/A”).

When the agent cannot tell two worlds apart from the value alone, that is *belief uncertainty*: it splits probability mass across the live worlds; it does not merge the boxes. Distinguishing **typo** from **wrong_product** is justified by *decision relevance*—they lead to different repairs (fix the digit vs. re-identify the product), so they must be separate states.

4 The belief model

4.1 Source-reliability prior

Before seeing evidence, the agent starts from a prior over the six worlds that depends only on *where the value came from*. We use three source tiers—manufacturer (MFR), mid-reliability retail

(MID), and sketchy aggregator (SKY)—each expressed as 100 tokens split over the worlds. The mid-reliability board is the anchor:

MID: $P(\text{correct}) = 0.50$, $\text{unit_error} = 0.10$, $\text{wrong_product} = 0.15$,
 $\text{typo} = 0.05$, $\text{missing} = 0.10$, $\text{garbled} = 0.10$.

MFR shifts mass toward **correct** (0.80); SKY away from it (0.30). These are *weakly-informative* priors: strong enough to rule out the absurd (via support ranges, a phone cannot have “100 ft of RAM”) but not so strong they override real evidence.

4.2 Likelihoods

The likelihood answers the backwards question: *if this world were true, how naturally would it produce this evidence signal?* Each $P(\text{signal} \mid \text{world})$ lives in $[0, 1]$ and the values *do not* sum to one across worlds—they are per-world sampling probabilities, not a distribution over worlds. The primary signal is “out-of-support” (oos); for a value like 160 GB RAM the likelihoods are $P(\text{oos} \mid \text{typo}) = 0.90$, $P(\text{oos} \mid \text{unit_error}) = 0.40$, $P(\text{oos} \mid \text{wrong_product}) = 0.10$, $P(\text{oos} \mid \text{correct}) = 0.02$, and 0.01 for *missing/garbled* (a parsed number is essentially never empty or junk).

Rigor note. For a binary support test the “in-support” case implicitly uses $1 - P(\text{oos} \mid \text{world})$. For any multi-valued signal (e.g. $\text{cross_source} \in \{\text{agree}, \text{disagree}, \text{absent}\}$) the conditional distribution must sum to one *over the evidence outcomes* for a fixed world, or the “absent” outcome silently leaks probability mass.

4.3 Bayesian update and chaining

The posterior is $\text{prior} \times \text{likelihood}$, renormalized:

$$P(\text{world} \mid e) = \frac{P(\text{world}) P(e \mid \text{world})}{\sum_w P(w) P(e \mid w)}. \quad (1)$$

On the 160 GB-RAM MID example, the single oos signal moves the board to $\text{typo} = 0.402$, $\text{unit_error} = 0.357$, $\text{wrong_product} = 0.134$, $P(\text{correct}) = 0.089$, and 0.009 each for the empty worlds. Note the base-rate discipline: **typo** lands at 0.40, not 0.95, precisely because its prior was low—striking evidence does not erase the prior.

Multiple clues chain: the posterior of one update becomes the prior of the next (Eq. 1 applied repeatedly). **This asserts conditional independence of the signals given the world**, $P(e_1, e_2 \mid w) = P(e_1 \mid w) P(e_2 \mid w)$, an assumption we state openly because it is questionable for our signals: a typo that pushes a value out of support will *also* tend to make it disagree with other sources—one latent cause, two signals—so chaining them double-counts and over-concentrates the posterior. This is the likely mechanistic root cause of the over-confidence we measure in §9. Principled fixes (a joint likelihood for correlated signal groups, or a naive-Bayes-style weight-of-evidence discount) are future work.

5 Assumptions, provenance, and the cost matrix

5.1 What is assumed vs. measured

The priors, likelihoods, and costs are **author-stated assumptions**, informed by experience browsing major B2C retail sites—not counts from a labelled dataset. We label them as such on purpose: a number without its provenance is “a guess wearing a lab coat.” The only *measured* quantities

in this paper are the evaluation results of §9, computed by running the model over labelled cases. Because the derived thresholds and posteriors are computed *from* the assumptions, they inherit the assumptions’ uncertainty: every “derived floor” or posterior below should be read as “given these stated assumptions.”

5.2 Asymmetric cost matrix

Not all mistakes cost the same; publishing a wrong value dominates. Costs are relative pain units, author-assigned on the principle “whatever ends up publishing wrong data is most expensive.” Table 1 gives cost(action, true world).

Table 1: Asymmetric cost matrix (author-stated assumption). Rows are actions, columns are the true world; lower is better. ACCEPT on any flaw is the worst outcome.

Action ↓ / World →	correct	unit_err	wrong_prod	typo	missing	garbled
ACCEPT	0	20	20	20	20	20
REPAIR	18	0	8	4	12	12
BUY	1	1	1	1	1	1
FLAG	2	2	2	2	2	2
REJECT	5	6	6	2	1	1

The asymmetry (20 for a wrong publish vs. 1–2 for buying/flagging) is what makes the problem interesting: even when correct is the single most likely world, a modest chance of a costly flaw can make ACCEPT the higher-cost choice.

6 Policy: minimum expected loss with an abstain option

6.1 Decision rule

For each action a , its expected loss under the current belief is

$$\mathbb{E}[\text{loss} \mid a] = \sum_{\text{world}} P(\text{world}) \text{cost}(a, \text{world}), \quad (2)$$

and the agent takes the action with the smallest expected loss. On the 160 GB posterior above, this yields $\mathbb{E}[\text{ACCEPT}] \approx 18.2$, $\mathbb{E}[\text{REPAIR}] \approx 4.5$, $\mathbb{E}[\text{REJECT}] \approx 4.0$, $\mathbb{E}[\text{FLAG}] = 2.0$, and $\mathbb{E}[\text{BUY}] = 1.0$: the engine’s answer is *don’t guess—buy more evidence*, a non-obvious decision produced by the model rather than by hand.

6.2 Derived accept threshold (Chow’s reject option)

ACCEPT (cost 20 if wrong, 0 if right) beats FLAG (flat 2) only when $(1 - P(\text{correct})) \times 20 < 2$, i.e. $P(\text{correct}) > 0.90$. So 90% is the cost-*derived* floor; we operate at a more conservative 95% for a publishing safety margin. Acting only above a cost-derived confidence and otherwise abstaining to a human is precisely Chow’s reject-option rule [2].

6.3 Buying information: a budget heuristic, not true VOI

Whether to BUY is governed by two rules. (i) A *value-of-information pre-filter*: if buying cannot change the action (e.g. missing/garbled dominate, or every plausible world already implies the

same action), do not buy. (ii) A *cost cap*: since FLAG costs 2 and guarantees a resolution, never spend more than 2 on buying—a hard cap of two BUYS, then FLAG. We are explicit that this is a **budget heuristic, not true value of information**: true VOI compares $\mathbb{E}[\text{loss now}]$ against $\mathbb{E}[\text{loss after the evidence}]$ and buys only if the expected reduction exceeds the fetch cost. The heuristic needs only a counter, no forecasting; its fully-correct form is a sequential/optimal-stopping problem [9], out of scope here. Section 7 replaces this heuristic with a true VOI rule and shows it makes a materially different decision on the worked case.

7 The information layer: bits, which clue to buy, and when to stop

The policy above answers *what to do given a belief*; it never asks *how much doubt remains* or *which evidence is worth buying*. This section adds that information layer on the same worked case (`ram_gb`= 160), reusing the posterior of §4 and the cost matrix of §5. No new assumptions are introduced beyond the per-clue likelihoods stated in Table 2; all numbers are reproduced by a standard-library script.

7.1 Measuring doubt in bits

We quantify total uncertainty with Shannon entropy $H(b) = -\sum_w b(w) \log_2 b(w)$, in bits. For six equally likely worlds $H = \log_2 6 = 2.585$ bits (the ceiling). The “mid”-reliability prior sits at $H = 2.123$ bits—already very unsure. After the single *out-of-support* clue the posterior

$$b_{\text{post}} = (0.089, 0.134, 0.357, 0.402, 0.009, 0.009)$$

has $H = 1.880$ bits, an *observed* information gain of only 0.243 bits.

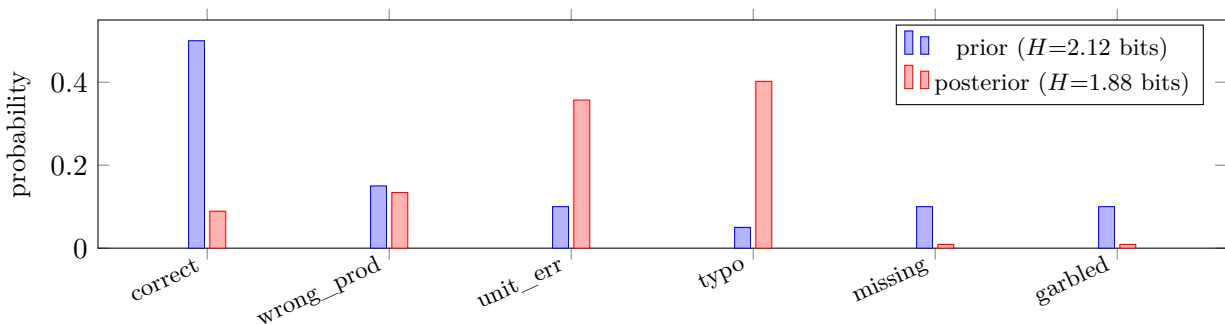


Figure 1: The out-of-support clue *flips the leader* (`correct`→`typo`) yet barely reduces entropy (−0.24 bits): changing the top-ranked world is not the same as reducing doubt.

7.2 Expected vs. observed gain

The 0.243 bits above is the gain *after* seeing `oos`=true. Before running the check the outcome is unknown: `oos`=true occurs with probability $P(\text{oos}) = \sum_w b(w)P(\text{oos} \mid w) = 0.112$ (then $H = 1.880$), and `oos`=false with probability 0.888 (then $H = 1.897$). The honest value of *running* the check is the conditional entropy $H(W \mid \text{clue}) = 0.112(1.880) + 0.888(1.897) = 1.895$ bits, giving an **expected information gain** of $2.123 - 1.895 = 0.228$ bits. Expected gain equals the mutual information between clue and hidden world; ranking clues by the *observed* number instead is a standard error, and we avoid it below.

7.3 Which clue to buy next: expected bits per unit cost

Three costed clues are available (a manufacturer spec-page proxy, a reference-dataset SKU lookup, and cross-source agreement). Table 2 states their two-outcome likelihoods (author assumptions) and the resulting expected gain and bits-per-cost, computed from the post-oos posterior.

Table 2: Ranking candidate clues on the worked case. The *most informative* clue is not the best *buy* once cost enters.

Clue	Cost	$\mathbb{E}[\text{gain}]$	Bits/cost	VOI
Manufacturer proxy = “16 GB”	2	0.250	0.125	0
Cross-source panel = “16 GB”	1	0.135	0.135	0
Reference catalogue (stale)	1	0.039	0.039	0

The manufacturer proxy carries the most raw information (0.250 bits) but the cross-source panel wins *per unit cost* (0.135 vs. 0.125)—the raw-bits and bits-per-cost rankings *disagree*. The reference catalogue is nearly worthless here (0.039 bits): with a scraped, stale catalogue a match barely discriminates the worlds. (Symmetrically, a *broad* cross-source panel would be the confounded one, reproducing a human critique we received (§13) that “five agreeing sites are not five independent votes”; here the author’s specific cross-source panel is curated and near-independent, while the available catalogue is the weak one.)

7.4 When to stop: a true value-of-information rule

Bits are not the stop criterion; expected *loss* is. We buy a clue iff

$$\text{VOI} = \mathbb{E}[\text{loss} \mid \text{best terminal action now}] - \mathbb{E}[\text{loss} \mid \text{best terminal action after the clue}] > \text{cost},$$

with the post-clue term averaged over the clue’s outcomes (re-selecting the cheapest terminal action in each branch). On the worked case the cheapest terminal action now is FLAG at $\mathbb{E}[\text{loss}] = 2.00$ (vs. REJECT 4.21, REPAIR 4.50, ACCEPT 18.22). Every clue—including the *most informative* one (manufacturer, 0.250 bits)—leaves FLAG cheapest in *both* branches, so $\mathbb{E}[\text{loss after}] = 2.00$ and $\text{VOI} = 0$ for all three (last column of Table 2): the agent should **flag now, not buy**—a different decision from the flat-cap heuristic of §6, which would have spent a buy first. The reason is the cost asymmetry: even a positive manufacturer result only lifts `unit_error` to ≈ 0.55 , and REPAIR on a 0.55 belief eats the cost-18 penalty $\approx 45\%$ of the time—worse than a guaranteed FLAG=2. VOI is not “never buy” (Table 3): a cheap confirming lookup becomes VOI-positive precisely where it can tip ACCEPT below FLAG.

Table 3: The VOI-positive band is thin: the rule refuses hopeless buys and spends only near the ACCEPT/FLAG frontier.

$P(\text{correct})$	Best terminal now	Net VOI of ref-lookup	Decision
0.40 (worked case)	FLAG @ 2.00	−1.00	flag now
0.86	FLAG @ 2.00	+0.12	buy
0.90	ACCEPT @ 2.00	+0.34	buy
0.92	ACCEPT @ 1.60	+0.05	buy

7.5 Two intuitions, tested honestly

(a) **Highest information $\neq$ best buy—confirmed, two ways.** The manufacturer clue is the most informative (0.250 bits) yet (i) loses on bits-per-cost to the cross-source panel ($0.125 < 0.135$) and (ii) has $\text{VOI} = 0$; information is not free. (b) **“Evidence can raise uncertainty”—not on our numbers.** In principle a single outcome can raise entropy, but we checked all three clues and every branch *lowers* it (manufacturer 1.59/1.68, cross-source 1.69/1.79, reference 1.84/1.84 vs. a 1.881-bit start). We decline to manufacture a row that would force the effect. It arises only when a clue’s outcome *contradicts the current front-runner*; here the front-runners are error worlds, so a “not 16” result is consistent with them and keeps concentrating. The durable lesson stands regardless: judge a clue by its *expected* effect, since gain is an average over outcomes.

7.6 What the information layer exposed (a falsified hypothesis)

Because FLAG is a flat, guaranteed 2, VOI-positive buys exist only in a thin band near the ACCEPT frontier (Table 3), so under our own matrix the engine should buy far less than the Week-1 cap assumed. A natural hypothesis is that this is merely because $\text{FLAG} = 2$ is cheap, so *raising* the human cost should revive buying. We tested it and it is **false**: as FLAG rises from 2 to 15, VOI rises off zero but *plateaus* (manufacturer 0.38, cross-source 0.23) and never clears the clue cost, because once $\text{FLAG} \geq 5$ the cheapest fallback switches to REJECT@4.21—capped by the matrix and independent of FLAG. On a high-entropy case, buying fails not because the human is cheap but because *no single clue can concentrate belief enough to escape into a cheaper action*. ($\text{FLAG} = 2$ flat-and-infallible is still an over-simplification, §11; the tested point is narrower: fixing it alone does not rescue buying on foggy cases.) The information analysis thus turned a buried modelling assumption into a visible, and falsifiable, driver of the buy policy—its main contribution here.

8 Experimental setup (methods)

Reproducibility. The model is transcribed verbatim into `src/domain.py` (the six worlds, the three priors, the likelihoods, the actions, the cost matrix, the 0.95 threshold, and the 2-buy cap) and executed by `src/agent.py`. Everything is pure Python 3.11 standard library—no third-party dependencies—so the results regenerate deterministically with `python3 generate_data.py` && `python3 evaluate.py`.

Data. We evaluate on $N = 40$ **labelled cases** (`data/cases.jsonl`), generated deterministically (seed 20260830). Fourteen are hand-authored “unit test” cases spanning all six worlds, all three sources, and every evidence signal, including two deliberate **wrong_product failure seeds** that look normal and even carry cross-source agreement. The remaining 26 are sampled: a world is drawn from a source’s prior, then each candidate signal fires with its per-world likelihood. **Labels are hidden at decision time**—the deciders see only source + value + evidence readings, and the true world is revealed afterward only to score, so the agent never marks its own paper.

The three deciders.

- **full** — the §6 policy: belief + expected loss + BUY (capped at 2) + FLAG.
- **autonomous** — the same belief and expected loss but *no human and no buy*; it must commit to ACCEPT/REPAIR/REJECT.
- **baseline** — a deliberately “dumb” calibrated score-plus-threshold policy: ACCEPT if $P(\text{correct}) \geq 0.90$, REJECT if $P(\text{correct}) \leq 0.40$, otherwise FLAG. It uses the *same* posterior $P(\text{correct})$ (so

the comparison is apples-to-apples) but ignores *which* failure world is likely, so it can never REPAIR or pick a targeted next probe. This is the honest strawman a practitioner suggested online (§13).

Metrics. Average decision cost (primary), human-review rate (fraction sent to FLAG), count of catastrophic wrong-publishes (ACCEPT on a flaw, cost 20), ACCEPT precision/recall, a reliability-by-confidence-bin calibration table, the fraction of cases where **full** and **baseline** pick the same action, and a paired Wilcoxon signed-rank test [10] on per-case cost differences to check whether the cost gap is more than sampling noise.

9 Results

9.1 Headline metrics

Table 4 summarizes the three policies over the 40 cases.

Table 4: Measured results on $N = 40$ labelled cases (labels hidden at decision time). Lower cost is better. All numbers are computed by `src/evaluate.py`.

Policy	Avg cost	Human-review rate	Wrong-publishes (cost 20)	ACCEPT prec.	ACCEPT recall
full	2.30	0.575	2	0.75	0.333
autonomous	3.075	0.000	2	0.818	0.500
baseline	3.10	0.325	2	0.75	0.333

The **full** policy has the lowest average cost, and it agrees with the **baseline** on only 62.5% of cases (25/40)—so the belief layer genuinely *changes* decisions; it is not score-plus-threshold with extra steps. But three observations temper the headline (§10).

9.2 Cost advantage: significant but narrow

The paired Wilcoxon signed-rank test on per-case cost differences finds the gap *is* statistically significant—**full vs. baseline** $p \approx 0.015$, **full vs. autonomous** $p \approx 0.003$ —but the **median per-case cost difference is 0**, and only **9 of 40** cases differ at all (all in **full**’s favor). The correct reading is therefore *real but narrow*: most cases tie, and the significant mean gap is driven by a minority of cheaper mid-cases plus heavy escalation, not by a broad per-case improvement. On $N = 40$ with a spiky cost distribution (a single cost-20 event shifts the mean by ~ 0.5), this is indicative, not definitive.

9.3 Calibration

Figure 2 plots, for the **full** policy, the mean predicted $P(\text{correct})$ against the empirical fraction that were truly correct, per confidence bin. The lower bins are well-behaved, but the *highest-stakes* bin ($P(\text{correct}) \geq 0.95$)—the bin on which ACCEPT fires—is only $\sim 60\%$ empirically correct.

Why this is load-bearing. The ACCEPT threshold in §6 was derived assuming $P(\text{correct})$ is the true probability of correct. If the true rate in the top bin is 0.60, then the *actual* expected loss of ACCEPT there is $(1 - 0.60) \times 20 = 8$, far above FLAG= 2 and REJECT= 5. In that case “ACCEPT

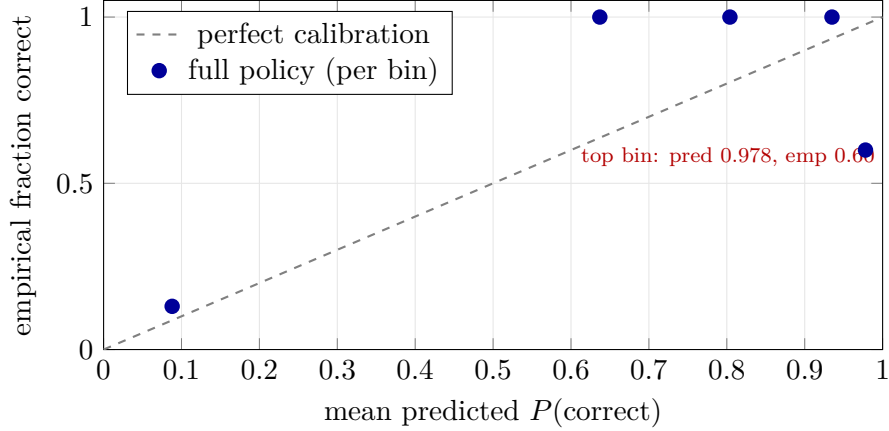


Figure 2: Reliability diagram for the **full** policy. Points on the dashed line are perfectly calibrated. The top bin sits far *below* the line—severe over-confidence exactly where a wrong ACCEPT costs 20. Caveat: that bin has only $n = 5$ (3/5 correct), so a 95% Wilson interval spans ≈ 0.23 – 0.88 ; we flag the *direction*, not a precise magnitude.

at $P(\text{correct}) \geq 0.95$ ” would be choosing the *worst* available action: the miscalibration does not merely add noise, it *inverts* the decision. The expected-loss policy is only sound once the posterior is calibrated; the 0.95 threshold is thus a *design target*, not a validated operational risk level.

9.4 Escalation and the shared blind spot

Figure 3 contrasts average cost with human-review rate. All three policies commit the *same* two catastrophic wrong-publishes: the two deliberate **wrong_product** seeds, which look normal and even carry cross-source agreement, are accepted by every policy. The belief layer does not prevent a single catastrophic error relative to the comparators; its lower average cost comes from routing 57.5% of cases to a human (vs. 32.5% baseline, 0% autonomous).

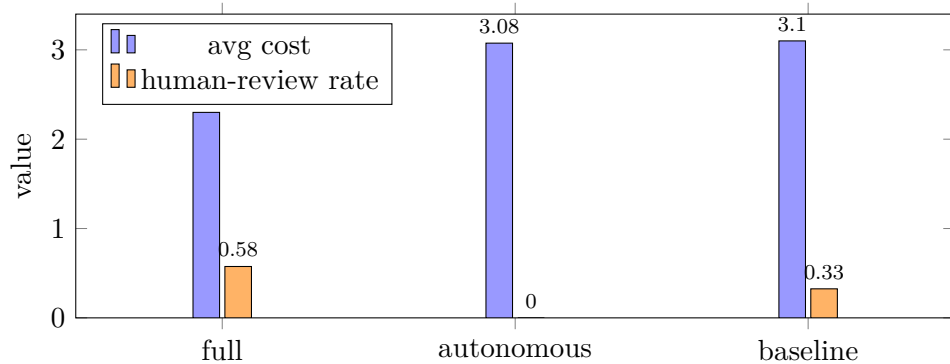


Figure 3: Average decision cost vs. human-review rate. **full** wins on cost but at the price of a 57.5% escalation rate; **autonomous** never escalates yet has higher ACCEPT precision (0.818 vs. 0.75). Wrong-publishes are identical (2/2/2, not shown), so the cost win is not a catastrophe-reduction win.

10 Does the belief layer “earn its keep”?

An earlier draft concluded that the belief layer “earns its keep” because it beats the baseline on cost. That conclusion is overstated, and we retract it in favor of a narrower, defensible one. Three facts, all from our own results, force the correction:

1. **Changing decisions $\neq$ better decisions.** Disagreeing with the baseline on 37.5% of cases shows the layer *differs*; it does not show it is *better*.
2. **The win is bought by escalation.** The lower average cost is largely a consequence of flagging 57.5% of cases at a flat FLAG cost of 2. A fair comparison must hold the human-review budget fixed—which we have not yet done. Indeed the **autonomous** policy, which never escalates, has *higher* ACCEPT precision (0.818 vs. 0.75).
3. **No catastrophe reduction.** All three policies eat the same two cost-20 wrong-publishes.

Honest conclusion. On a preliminary 40-case sample the full policy showed a lower average decision cost (2.30 vs. 3.10; paired Wilcoxon $p \approx 0.015$ but median per-case difference 0), primarily by routing 57.5% of cases to human review; it prevents no more catastrophic wrong-publishes than the baseline. What *is* supported: the belief layer changes decisions, and—unlike the baseline—it can REPAIR and pick targeted probes, which are qualitatively different capabilities that a fixed-escalation-budget experiment on a larger dataset would be needed to price.

11 Failure modes and limitations

Failure modes (ranked).

1. **wrong_product blind spot (highest severity).** A value that is in-support *and* agrees with sources but is the wrong variant is accepted (cost 20). Value + support + agreement cannot catch this; only a manufacturer proxy or a human can. Demonstrated by the two failure seeds.
2. **Coordinated cross-source error.** The agreement signal assumes source independence, which shared upstream feeds violate; five copies of one bad feed read as strong (false) agreement.
3. **Schema/key mismatch.** If the upstream step mislabels which field a value belongs to, the agent cannot recover it (out of scope by assumption).
4. **Mislabeled source reliability.** A “manufacturer” source that is actually a reseller corrupts the prior.
5. **Stale support ranges.** New hardware outside the current support range is wrongly flagged/rejected (cold-start).

Honest “what it cannot do.” (i) It cannot verify images (pure human-only evidence). (ii) Its priors are assumed, not counted—there is no general way to prove a strong prior is justified. (iii) It cannot detect coordinated cross-source errors. (iv) Support ranges need ongoing maintenance. (v) It assumes correct upstream field identification. (vi) The evaluation is active-learning-biased (§12). (vii) FLAG is modelled as a flat cost 2 that *perfectly* resolves the case; a real reviewer has an error rate, latency, and a per-item cost that grows with volume, so the true cost of the full policy’s heavy escalation is understated. (viii) The Bayesian outputs are *conditional on* the assumed priors/likelihoods/costs and would move if those were re-estimated from labelled data.

Roadmap. The path to production, out of scope for Week 1: an identity/variant resolution subsystem (the real fix for the **wrong_product** blind spot); an evidence-lineage/provenance graph

or an effective-sample-size discount $k/(1 + (k - 1)\rho)$ to replace the hand-tuned agreement discount; loss-aware VOI; temporal/version/region semantics; an independently-audited gold set including auto-accepts; risk-tiering of fields and per-product (not per-field) human review; and recalibration (Platt/temperature/isotonic) before any threshold is used operationally.

12 Threats to validity

- **Small sample.** $N = 40$ with no confidence intervals; the cost gap is significant by a paired test but narrow (median difference 0). Conclusions are preliminary.
- **Active-learning evaluation bias.** The agent auto-accepts a bucket it never verifies and only humans see the flagged bucket, so scoring across the labelled set over-weights the hard cases and the calibration numbers themselves are measured on a biased sample. The honest fix is to randomly audit a slice of the auto-accepted bucket and importance-weight it back.
- **Synthetic data.** Cases are generated from the same likelihood model the agent uses, which flatters the agent; only the 14 hand-authored cases and the deliberate failure seeds inject independent structure.
- **Assumptions drive the numbers.** Because priors/likelihoods/costs are assumed, the derived thresholds are internally consistent but not empirically validated.
- **Calibration magnitude.** The “60% in the top bin” is 3/5; the direction (over-confidence) is clear, the magnitude is not established at $n = 5$.

13 AI-use statement, human discussions, and conclusion

AI-use statement. This deliverable was built collaboratively with an AI coding assistant (Claude, via the Cursor harness) used as a Socratic pair: it taught the underlying concepts, asked clarifying questions, and drafted files, while the author owned every design decision (the six worlds, the prior token splits, the cost ordering, the thresholds, and the buy-stop rule). All conversations, decisions, and their justifications are logged in `WORKING-CONTEXT.md`. Three *adversarial* AI reviews were run in separate sessions across different models—a practitioner review (ChatGPT), a probability/statistics review (Claude), and a preprint/writing review (Gemini)—and each comment was adjudicated (accept/reject with a reason) in `review-record.md`. The probability review validated the core Bayesian mechanics and surfaced the conditional-independence and VOI-vs-budget issues; the preprint and practitioner reviews forced the calibration and “earns its keep” honesty corrections made throughout this paper; the paired significance test was added in direct response to a reviewer’s challenge.

Human discussions. The design was also shaped by five real online discussions (documented in `discussion-record.md`). A web-scraping practitioner’s point—that agreement among sources copying the same feed is not independence—became the correlated-evidence discount and the `wrong_product`/F2 failure analysis; a machine-learning practitioner’s “build the dumb score-plus-threshold baseline and check if the belief layer pays rent” became the **baseline** decider and the agreement test; and a statistician’s caution about justifying strong priors became the weakly-informative framing and the label-free prior-predictive calibration idea.

Conclusion. We presented an explicit, auditable belief-engine decision model for normalizing scraped product specifications, and we evaluated it honestly. The framework is sound and the empirical cost advantage is statistically real, but it is narrow, it is bought by escalation rather

than by catching the expensive errors, and it rests on a posterior that is not yet calibrated in the one bin that matters. The contribution is therefore a disciplined *way of thinking*—priors with provenance, asymmetric costs, minimum expected loss, a principled abstain, and a refusal to dress assumptions up as measurements—rather than a production-ready system. The information layer (§7) sharpens rather than softens this honesty: it shows that choosing evidence by expected bits-per-cost reorders our clues, that a highly informative clue can still be the wrong *buy* (or even raise uncertainty on some outcomes), and that under our own cost matrix the value of information is usually zero because guaranteed human review is cheap—so the buy policy was quietly resting on an optimistic flat human cost. The most valuable next step is not more machinery but better evidence: a calibrated posterior on an independently-audited sample, a fixed-escalation-budget comparison at larger scale, and a cost model for the human that does not treat review as free and infallible.

A Required core questions

Answers in the author’s own words, grounded in this problem wherever possible.

1. **Hidden state, and ours?** A fact the agent cannot observe at decision time but needs in order to act. Ours is *which of six worlds* produced a scraped value—correct, unit_error, wrong_product, typo, missing, garbled—since the true canonical spec is never seen.
2. **Why is directly predicting an answer insufficient?** A point prediction (“it’s a typo”) throws away how *sure* we are and which alternatives are live, which is exactly what an asymmetric cost matrix needs. Predicting “correct” with 55% confidence and with 95% confidence should lead to different actions; a bare label cannot express that.
3. **Belief distribution, and why sum to one?** A probability over all hidden worlds representing our current uncertainty. It sums to one because the worlds are exhaustive (§3)—the true value is *some* world, so the total probability of “it is one of them” must be 1.
4. **Prior vs. posterior?** The prior is the belief before this field’s clues (our source-reliability board, e.g. $P(\text{correct}) = 0.50$ on a mid source); the posterior is the belief after applying a clue via Bayes (e.g. $P(\text{correct}) = 0.089$ after the out-of-support signal).
5. **Likelihood vs. probability?** Likelihood is $P(\text{clue} \mid \text{world})$ —how naturally each world would *produce* the observed clue—read as a function of the world for fixed evidence, so it need not sum to one across worlds. A probability (e.g. the posterior) is normalized over worlds.
6. **What does agent uncertainty mean?** That its belief mass is spread over several worlds rather than concentrated on one; operationally, that no single terminal action is clearly cheapest. Our post-oos belief is uncertain: three error worlds remain live.
7. **Entropy, and why it is uncertainty?** $H = -\sum_w b(w) \log_2 b(w)$: the average number of yes/no questions needed to pin down the world. It is maximal when belief is uniform (maximum confusion) and zero when one world has all the mass (certainty), so it *is* a measure of doubt.
8. **A bit, plainly?** One yes/no question’s worth of uncertainty—the doubt in a fair coin flip. Eight equally likely options are three bits because three good halving questions identify the answer.

9. **Information gain vs. entropy?** Information gain is the drop in entropy from receiving evidence. *Expected* gain = $H(\text{prior}) - H(W \mid \text{clue})$; on our case the out-of-support clue’s expected gain is 0.228 bits.
10. **Conditional entropy?** $H(W \mid \text{clue})$: the entropy we *expect* to remain after a clue, averaged over its possible outcomes weighted by their probabilities ($0.112 \cdot 1.880 + 0.888 \cdot 1.897 = 1.895$ bits here). Information gain is the starting entropy minus this.
11. **Mutual information vs. correlation?** Mutual information is the expected information one variable gives about another (identical to expected information gain); it is zero only under independence and captures *any* dependence. Correlation only captures *linear* co-movement, so a variable and its square can have zero correlation yet full mutual information.
12. **KL divergence, and why not a distance?** $KL(p \parallel q) = \sum p \log_2 \frac{p}{q}$ measures the extra bits paid for coding samples from p using a code built for q . It is not a distance because it is asymmetric and violates the triangle inequality; it is zero iff $p = q$.
13. **Why KL asymmetric? Small example?** Because it weights the gap by p , the “true” side. With $p = (0.9, 0.1)$ and $q = (0.5, 0.5)$, $KL(p \parallel q) \approx 0.53$ bits, but $KL(q \parallel p) \approx 0.74$ bits—swapping the arguments changes the penalty, since regions where p is large but q is small hurt most.
14. **Jensen–Shannon, and why introduced?** $JS(p, q) = \frac{1}{2}KL(p \parallel m) + \frac{1}{2}KL(q \parallel m)$ with $m = \frac{1}{2}(p + q)$. It was introduced to fix KL’s flaws: it is symmetric, always finite (no divide-by-zero when one distribution has zero mass), and its square root is a true metric—useful for comparing, e.g., two sources’ value distributions for the same field.
15. **Calibration, and why it matters?** An agent is calibrated if things it calls 90%-likely happen about 90% of the time. It matters because our expected-loss math trusts the numbers: our top confidence bin is only ~60% correct, which (if real) inverts the ACCEPT decision the threshold was meant to license (§9).
16. **Expected cost, and how we derived the threshold?** Expected cost of an action = $\sum_w b(w) C(\text{action}, w)$. Since ACCEPT costs 20 if wrong and FLAG a flat 2, ACCEPT wins only when $(1 - P(\text{correct})) \cdot 20 < 2$, i.e. $P(\text{correct}) > 0.90$ —so 90% is the cost-*derived* floor (we operate at 95% for margin).
17. **Value of information; when is more not worth it?** VOI is the expected reduction in decision loss from a clue, and you buy only if VOI exceeds the clue’s cost. It is not worth it when no outcome would change the chosen action—on our worked case FLAG at 2 dominates in every branch, so VOI= 0 and the agent should flag, not buy (§7).
18. **Can a question be informative yet useless? Our example.** Yes. The manufacturer proxy is our *most* informative clue (0.250 bits) yet has VOI= 0 on the 160 GB case, because whatever it returns, FLAG stays cheapest—informative, but it changes no action, so it is useless *there*.
19. **Distribution shift, and detection?** When the world the model was tuned on stops matching reality (a new category, a new copying pattern). Our agent could detect it by prior-predictive monitoring—if the observed rate of out-of-support values, empties, or cross-source disagreement drifts from what the priors/likelihoods predict (the $P(\text{clue})$ normalizer), the assumptions are stale.

20. **When to stop and act? The rule.** Act when any fires: (i) one terminal action’s expected loss is clearly lowest (belief is decisive); (ii) no affordable clue is VOI-positive; (iii) two buys stop moving the posterior (correlated evidence); otherwise buy the highest positive- $(\text{VOI} - \text{cost})$ clue and re-evaluate (§7).

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